IIPC Assistant

AI-Powered Web Archiving Platform

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Contents

Summary

The IIPC Assistant is an AI-powered platform designed to enhance access to International Internet Preservation Consortium (IIPC) materials by enabling natural-language querying of web archiving resources. It integrates vector embeddings (BAAI/bge-base-en-v1.5) with generative AI (Google Gemini) to provide context-aware answers supported by verifiable source references.

The system successfully processed 587 IIPC conference posters, presentations, and transcript videos, transforming them into a searchable knowledge base. Development began with harvesting metadata via the OAI-PMH API, followed by text extraction and AI-enhanced restructuring to produce coherent, semantically meaningful content for retrieval.

A Retrieval-Augmented Generation (RAG) approach was adopted, utilizing FAISS vector database for similarity search and an improved retrieval method that groups chunks by document ID and ranks entire documents by their best chunk. The system is built with a React + TypeScript frontend, Flask backend, and delivers a responsive, cross-device interface.

Through iterative improvements in dataset preparation, text enhancement, and retrieval optimization, the IIPC Assistant now produces accurate answers and significantly enhances the discoverability and research value of IIPC collections.

1 Problem

The primary challenge addressed by this project is the limited accessibility and discoverability of specialized web archiving resources from the International Internet Preservation Consortium (IIPC). Traditional search methods often fail to provide context-aware, natural language querying capabilities, making it difficult for researchers, practitioners, and students to efficiently explore conference materials, posters , and presentations.

Key issues include:

- Fragmented access to historical web archiving knowledge spanning multiple years
- Poor handling of unstructured or poorly formatted extracted text from PDFs and videos
- Lack of semantic understanding in search, leading to irrelevant or incomplete results
- Absence of intelligent synthesis across multiple documents

These problems result in underutilization of valuable IIPC resources, hindering research in digital preservation and web archiving fields.

2 Aim

The primary aim of this project is to develop an AI-powered platform that enhances the accessibility and usability of IIPC materials through intelligent search and natural language interaction. Specific objectives include:

- Create a comprehensive knowledge base from 587+ IIPC documents with preserved metadata
- Implement semantic search using advanced vector embeddings for accurate retrieval
- Develop context-aware response generation grounded in source materials
- Provide a modern, responsive web interface for seamless user experience
- Facilitate research in web archiving by enabling natural language queries

The system aims to bridge the gap between traditional digital libraries and modern AI-driven knowledge discovery tools.

3 Survey of Existing Similar Idea

Several existing platforms and tools address aspects of digital preservation and intelligent search in archival contexts:

- **UNT Digital Library**: Provides basic search for IIPC materials but lacks semantic understanding and natural language processing.
- **Google Scholar with AI Overviews**: Provides academic search with AI summaries but not specialized for web archiving.
- **Perplexity AI**: General knowledge search with sources but lacks domain-specific focus on IIPC materials.
- **WebArchive Discovery**: Indexing tool for WARC files but limited user interface.

These solutions demonstrate varying levels of AI integration but none provide specialized, context-aware access to IIPC conference materials with verifiable sourcing.

4 Proposed Solution

The proposed solution is a Retrieval-Augmented Generation (RAG) platform specialized for IIPC materials:

- Data harvesting via OAI-PMH with multi-format extraction
- AI-enhanced text restructuring for improved semantics
- Vector database for efficient similarity search
- Generative AI for context-aware responses
- Modern web interface with real-time chat

This addresses identified gaps by providing domain-specific, verifiable AI assistance.

5 Model Architecture

The core architecture integrates:

- Embedding Model: BAAI/bge-base-en-v1.5 (768 dimensions)
- Vector Store: FAISS with L2 indexing
- LLM: Google Gemini 2.5 Flash-Lite
- Frontend: React with real-time updates
- Backend: Flask with API endpoints

See Figure 2 for detailed RAG flow.

6 Tools

Key tools and technologies:

- **Data Processing**: Jupyter Notebooks, PyMuPDF, Tesseract OCR
- **AI/ML**: Hugging Face Transformers, FAISS, Google Gemini API
- **Database**: Supabase PostgreSQL
- **Backend**: Flask
- **Frontend**: React, TypeScript, Tailwind CSS, Vite

7 User Requirements

The IIPC Assistant is designed to meet the needs of researchers, archivists, and digital preservationists accessing International Internet Preservation Consortium (IIPC) materials. The following requirements ensure the system is intuitive, effective, and accessible to its target audience.

7.1 End-User Needs

- **Semantic Search**: Users require the ability to query the system using natural language to retrieve relevant IIPC materials. The system must understand query intent and return contextually accurate results, leveraging semantic embeddings (BAAI/bge-base-en-v1.5) and FAISS for similarity search.
- **AI-Powered Responses**: Users expect context-aware answers generated from retrieved documents, supported by Google Gemini.
- **Filtering Capabilities**: Users need to filter search results by metadata such as publication year, document type (e.g., PDF, presentation, video transcript), and subject to narrow down relevant content efficiently.
- **Browsing and Exploration**: Users require a browsable interface to explore the IIPC collection, including access to metadata (title, creator, date) and full-text content stored in the Supabase database.
- **Cross-Device Access**: The system must support access from desktops, tablets, and mobile devices, ensuring seamless interaction for users in diverse research environments.

7.2 Features to Support Usability and Accessibility

- **Responsive Design**: The React-based frontend, styled with Tailwind CSS, ensures a consistent and intuitive user experience across devices, with adaptive layouts for varying screen sizes.
- **User-Friendly Query Input**: A natural-language input field allows users to pose questions without requiring technical knowledge of search syntax, supported by TypeScript for robust frontend interactions.
- **Result Visualization**: Search results are presented with clear metadata previews (e.g., title, creator, date) and snippets of relevant content, enhancing usability for quick evaluation.

8 System Requirements

The IIPC Assistant requires some specific functional requirements

8.1 Functional Requirements

The system must perform the following core functions:

- **Data Harvesting**: Automatically extract metadata and content from 587 IIPC documents via the OAI-PMH API, supporting PDF text extraction (PyMuPDF), OCR fallback (Tesseract), and VTT transcript processing for videos.
- **Text Processing and Enhancement**: Clean and restructure extracted text using Google Gemini 2.5 Flash-Lite to produce coherent, semantically meaningful content, preserving technical terminology and context.
- **Semantic Search**: Implement FAISS-based vector search with 768-dimensional embeddings from BAAI/bge-base-en-v1.5, enabling retrieval of relevant document chunks based on natural-language queries.
- **Retrieval-Augmented Generation (RAG)**: Combine FAISS retrieval with Gemini-generated responses, ensuring answers are grounded in retrieved content with minimal hallucination.
- **Database Management**: Store metadata (title, creator, date, description, subject, item type, ARK URL) and content in a Supabase PostgreSQL database with real-time query capabilities.
- **User Interface**: Provide a React-based frontend with TypeScript for type-safe interactions, supporting natural-language queries, result filtering, and metadata browsing.

9 Introduction

9.1 Project Overview

The IIPC Assistant is an AI-powered platform designed to enhance access to International Internet Preservation Consortium (IIPC) materials by enabling natural-language querying of web archiving resources. The platform integrates vector embeddings with generative AI to provide context-aware answers supported by verifiable source references.

9.2 System Architecture

The system employs a hybrid approach combining Supabase for structured data and FAISS for vector search. The architecture consists of:

- React frontend with TypeScript for type safety
- Flask backend
- FAISS vector database for efficient similarity search
- Google Gemini AI for intelligent response generation
- BAAI/bge-base-en-v1.5 for high-quality text embeddings
- Supabase for PostgreSQL database and real-time features

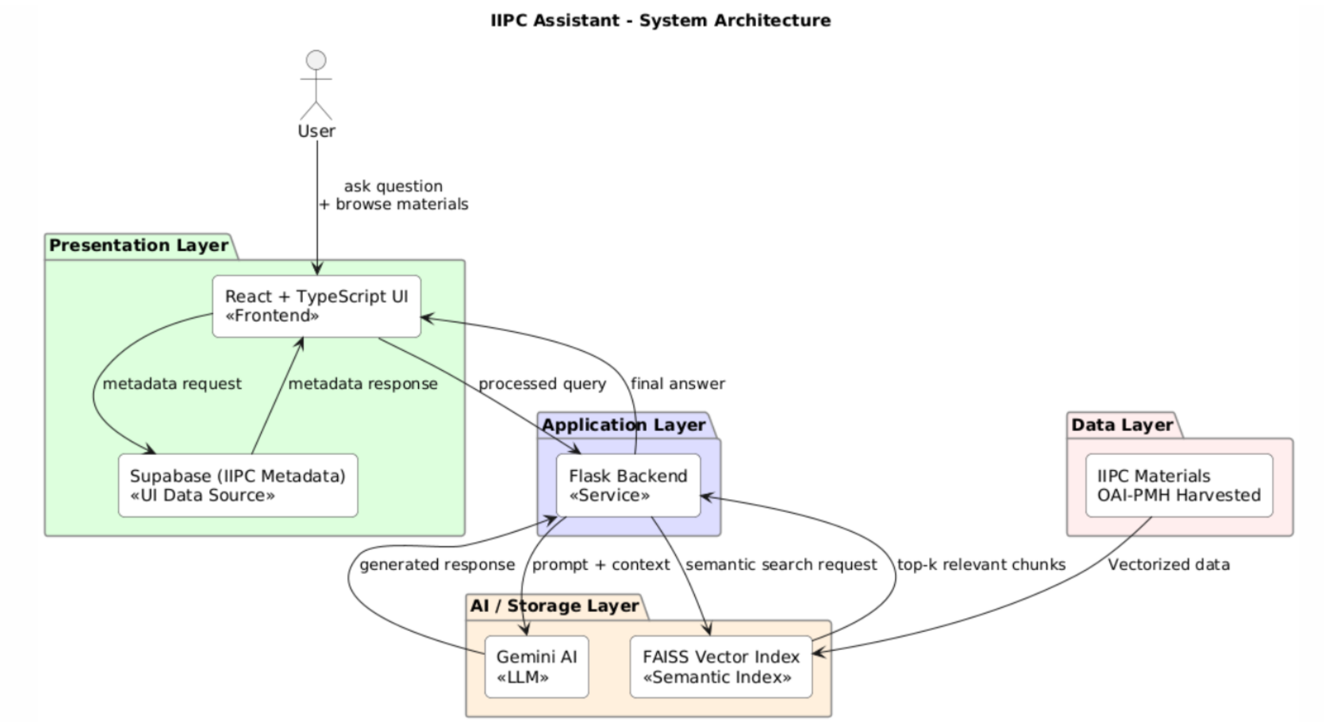

Figure 1: Overall system architecture of the IIPC Assistant platform. Frontend, backend, vector search, and AI services are shown.

10 Methodology

10.1 Data Pipeline Development

The project includes a comprehensive data processing pipeline from raw IIPC materials to the final intelligent chatbot system, implemented through three main Jupyter notebooks.

10.1.1 Data Harvesting Pipeline

The automated extraction of IIPC conference materials from UNT Digital Library was implemented with the following capabilities:

- **OAI-PMH Protocol Integration**: Harvests metadata using Open Archives Initiative standards
- **Multi-format Content Processing**: PDF text extraction with PyMuPDF and OCR fallback (Tesseract), VTT transcript extraction for video materials, intelligent file type detection and routing
- **Robust Data Collection**: ARK URL persistence for long-term access, complete metadata capture (title, creator, subject, description, dates), error handling for network failures and malformed data

The harvesting process successfully collected 587 IIPC documents with full metadata and extracted content.

10.1.2 AI-Powered Text Enhancement

Raw extracted text was transformed into structured, meaningful content using advanced AI through the following pipeline:

1. **Primary Text Cleaning**: Normalization of line breaks and character encoding, removal of OCR artifacts and formatting noise, standardization of whitespace and punctuation
2. **AI-Powered Restructuring**: Google Gemini 2.5 Flash-Lite integration for intelligent text processing, conversion of slide presentations into readable narrative format, preservation of technical terminology and context, rate-limited processing (15 RPM) for API compliance
3. **Quality Assurance**: Removal of AI-generated formatting artifacts, content validation and integrity checks, final cleanup of markdown and special characters

Performance metrics achieved:

- Processing rate: 4.1 seconds per document (API rate limit compliance)
- Quality improvement: Structured presentations from raw OCR text

10.2 Semantic Search and Chatbot Development

The intelligent question-answering system was developed with semantic search capabilities using an advanced NLP architecture:

10.2.1 Embedding Generation

- **State-of-the-art Embeddings**: BAAI/bge-base-en-v1.5 model for superior semantic understanding
- **Intelligent Text Chunking**: Token-aware chunking (450 tokens per chunk), 50-token overlap for context preservation, handles long documents exceeding model limits
- **Metadata Enrichment**: Combines title, creator, subject, and content for comprehensive context

10.2.2 Vector Database Implementation

The system uses FAISS for efficient vector similarity search with the following specifications:

- Vector Dimensions: 768-dimensional embeddings for rich semantic representation
- Accuracy: High precision through semantic similarity matching
- Scalability: Efficient handling of large document collections

10.3 Retrieval-Augmented Generation (RAG) System

The RAG system implements:

- **Semantic Retrieval**: FAISS-powered similarity search for relevant document chunks
- **Context-Aware Generation**: Gemini AI provides responses grounded in retrieved content
- **Domain Expertise**: Specialized knowledge in web archiving, digital preservation, and IIPC topics

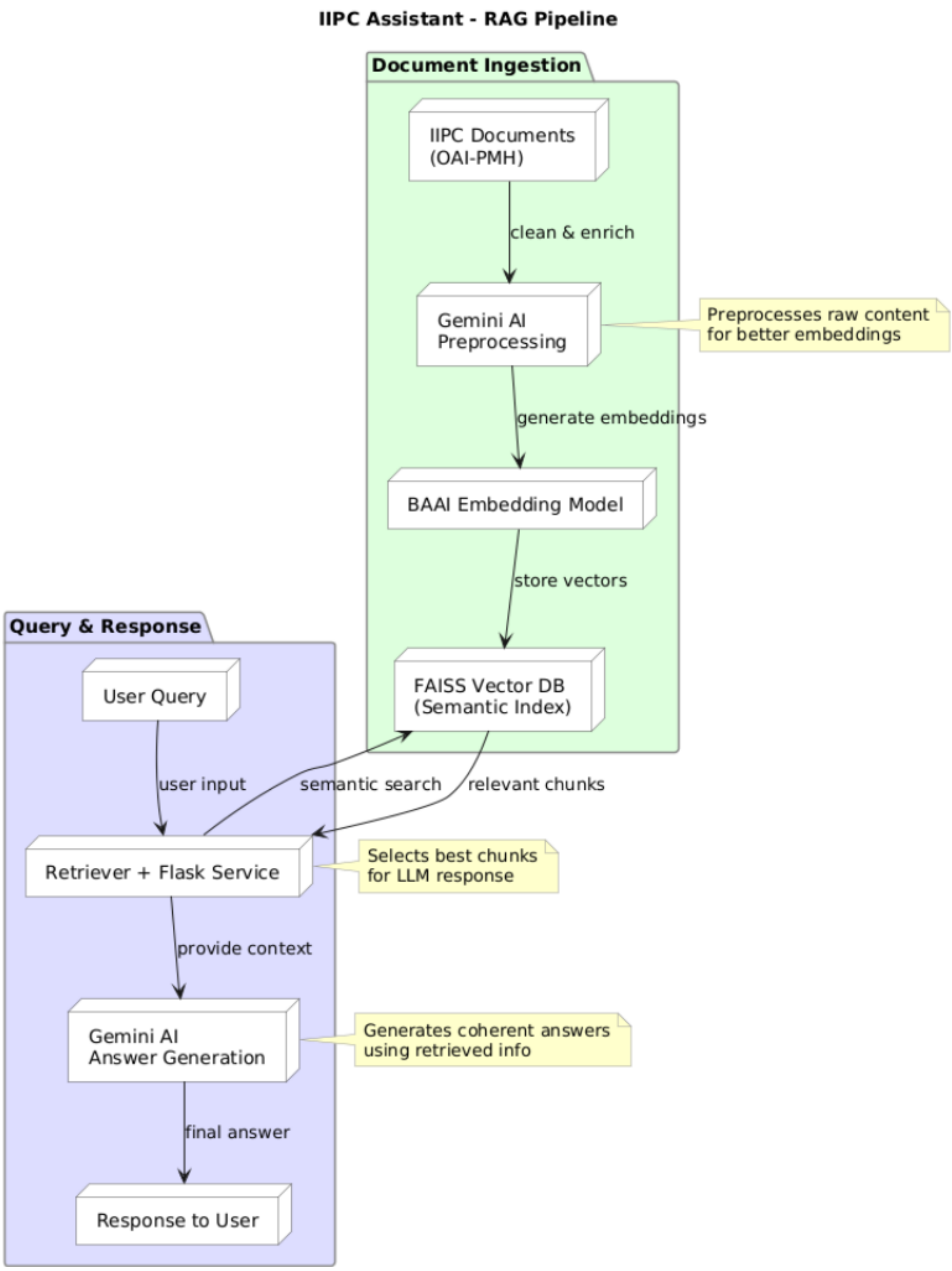

Figure 2: Retrieval-Augmented Generation (RAG) architecture: chunking, FAISS retrieval, document re-ranking, and Gemini-based answer generation.

11 Use Case Diagram

11.1 Actors

- **User** — as *Researcher / Student.*
- **External systems**:
  - Supabase Metadata Store (database)
  - FAISS Vector Index (database)
  - Google Gemini (AI Model)

11.2 Main Use Cases

- **Browse Conference Materials** — view available documents/records.
- **Apply Filters to Results** — narrow browse/search results .
- **Get AI-Powered Answer** — submit a natural-language query and receive a generated answer (interacts with FAISS and Gemini per UML).

11.3 Diagram

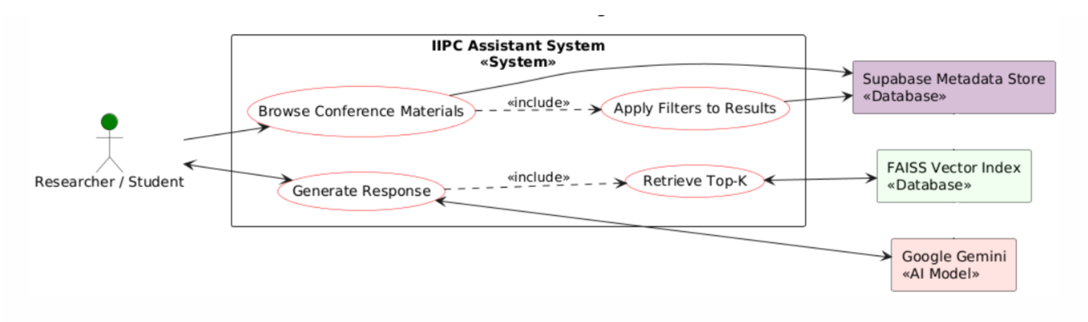

Figure 3: Use Case Diagram: IIPC Assistant

11.4 functional testing

- **Test 1 — Browse Materials:** Open the Browse view; expected result — list of conference materials is displayed (title/metadata previews).
- **Test 2 — Apply Filters:** From Browse, apply a simple filter (e.g., by year or item type); expected result — results are narrowed accordingly and still link to metadata store.
- **Test 3 — Get AI-Powered Answer:** Submit a query to the Chat/Answer flow; expected result — backend queries FAISS and Gemini and returns an answer with associated source references.

12 Sequence Diagram

12.1 Workflow

Below are concise workflow comments that reflect the exact flow in your provided sequence diagram:

1. **User → Frontend UI:** user enters a natural-language query (example shown: "Difference between crawling and harvesting?").
2. **Frontend UI → Backend API:** frontend sends a POST request to `/api/chat` containing the query.
3. **Backend:** generates an embedding for the query.
4. **Backend → Vector DB (FAISS):** backend searches the FAISS index with the embedding requesting top k chunks (k=30 in diagram).
5. **Vector DB → Backend:** FAISS returns the top 30 chunks.
6. **Backend internal step (noted in diagram):** improved retrieval method — group chunks by document ID, rank documents by best-chunk score, select top 3 documents, then take top 10 chunks from those documents.
7. **Backend → LLM (Gemini):** backend constructs a prompt with the query and retrieved context, then sends it to the LLM for completion.
8. **LLM → Backend:** LLM returns the generated answer.
9. **Backend → Frontend:** backend returns JSON containing the answer.
10. **Frontend → User:** frontend displays the answer.

12.2 Diagram

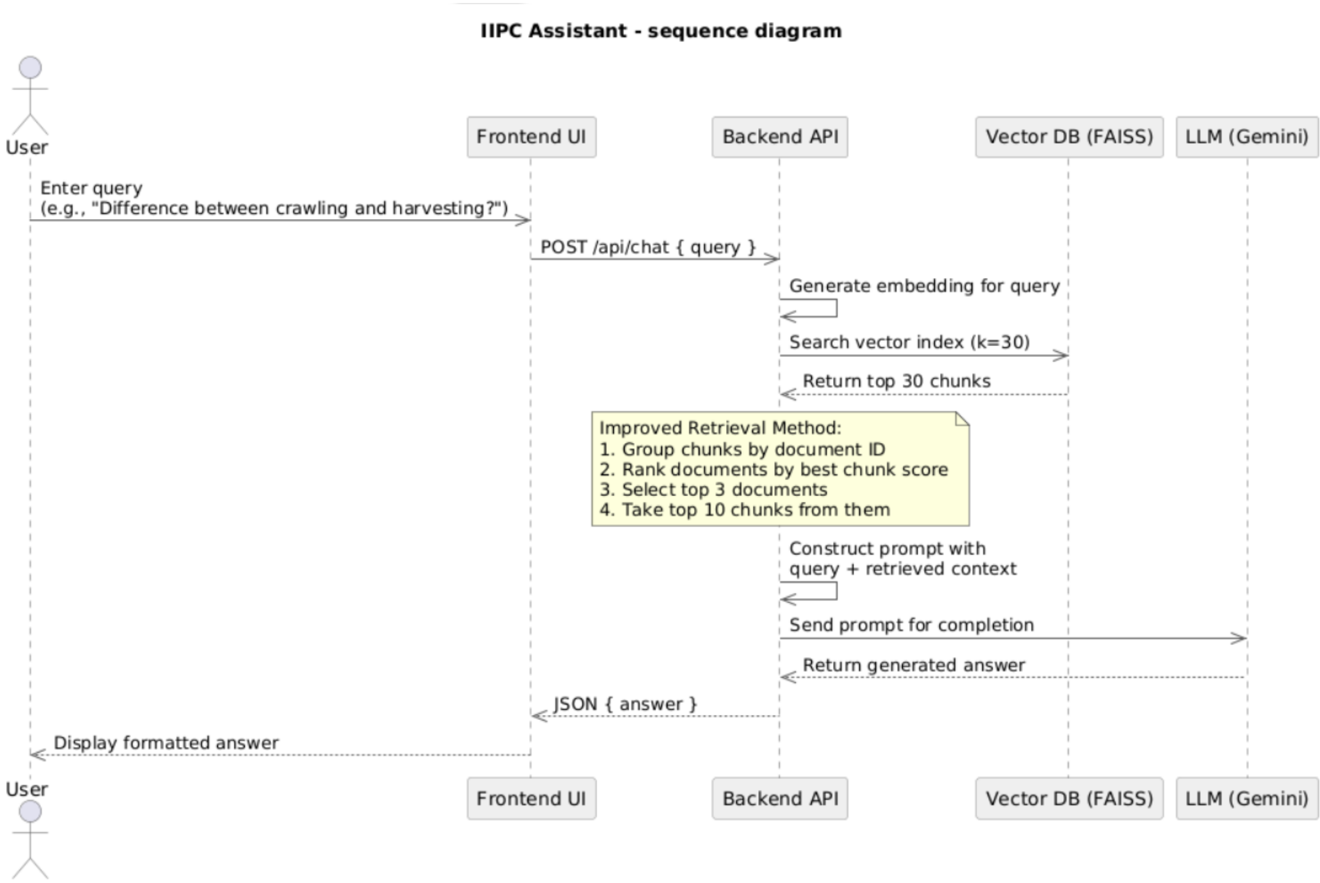

Figure 4: Sequence Diagram: Query flow for IIPC Assistant

13 Results

13.1 Data Processing Metrics

The complete data pipeline achieved the following performance metrics:

Table 1: Pipeline Performance Metrics

Stage	Input	Output
Harvesting	OAI-PMH Endpoint	587 Raw Documents
AI Enhancement	Raw Text	Structured Content
Embedding Generation	Enhanced Text	Vector Database
Chatbot Integration	User Queries	AI Responses

13.2 System Architecture Components

The final system architecture includes:

- Frontend: React with TypeScript, Tailwind CSS, Vite build system
- Backend: Flask web framework, FAISS vector database, Google Gemini AI integration
- Database: Supabase PostgreSQL with real-time capabilities
- Embedding Model: BAAI/bge-base-en-v1.5 (768 dimensions)

13.3 Database Schema

The structured data is stored in Supabase with the following schema:

```
CREATE TABLE iipc_materials (
  id SERIAL PRIMARY KEY,
  title TEXT,
  creator TEXT,
  date DATE,
  description TEXT,
  subject TEXT,
  item_type TEXT,
  ark_url TEXT,
  content TEXT,
  created_at TIMESTAMP DEFAULT NOW()
);
```

Listing 1: Database Schema

13.4 Technical Specifications

13.4.1 Multi-Modal Content Processing

- PDF documents with complex layouts and technical diagrams
- Video presentations with synchronized transcripts

13.4.2 AI Integration Strategy

- Text Enhancement: Gemini 2.5 Flash-Lite for content restructuring
- Response Generation: Context-aware answers with hallucination prevention
- Embedding Quality: BAAI BGE model for superior semantic understanding

14 Discussion

14.1 Design Decisions and Rationale

14.1.1 RAG Architecture Selection

The decision to implement a Retrieval-Augmented Generation (RAG) architecture rather than fine-tuning a language model was guided by the characteristics of the dataset. The collected materials — comprising posters, presentations, and transcripts — were heterogeneous and lacked a consistent question–answer structure. Fine-tuning typically requires well-labeled training pairs of input prompts and corresponding target responses, which would have necessitated extensive preprocessing and manual curation to prepare the dataset for supervised training.

RAG decouples information retrieval from response generation. Instead of embedding knowledge directly into the model weights, it dynamically retrieves relevant context at query time using an external index such as FAISS. This approach enables rapid integration of new or updated content without retraining, enhances transparency by exposing the original source documents, and preserves domain-specific accuracy. Furthermore, RAG is inherently well-suited to handling long-form, unstructured content, as it grounds responses in retrieved passages rather than relying solely on patterns memorized during model training.

14.1.2 Text Enhancement Strategy

The initial extracted text was poorly structured, resulting in inaccurate or inconsistent answers. This critical issue was addressed by employing Gemini AI to restructure and enrich the text, producing coherent and semantically meaningful input for retrieval. This preprocessing step was essential for system performance.

```
Mark Phillips, UNT Libraries \nAbbie Grotke, Library of Congress \n@vphill |
mark.phillips@unt.edu\n@agrotke | abgr@loc.gov\nThe End Of Term Archive:
Collaboratively Preserving the United States \nGovernment Web\nNovember 14, 2018\n
\nit all began a long, long, time ago, in a far away place
\nhttps://flic.kr/p/4N2jHU\nhttps://flic.kr/p/4JNkLE\n(to the United States, not
here)\n2\n\noriginal end of term web archive partners\nfor 2008/2012/2016 - all
IIPC & NDIIPP/NDSA partners\nNovember 14, 2018\nIIPC - WAC 2018\n3\n\nextant gov
web archiving efforts\nCapture, Preservation, & Access\n•\nLOC: .gov, election,
other\n•\nGPO: agency sites, often ephemeral \n•\nNARA: congressional web harvest
every \n2 years\n•\nIA: global & curated crawls\n•\nAgency-level: NIH/NLM, DOE,
DOL, HHS, \nCMS, others, using Archive-It or other \ntools\n•\nUNT & Others:
Topical .gov collecting\n•\nInternal agency guidelines\nCommunity Efforts\n•
\nFederal Web Archiving Group\n•\nmost of those at left plus other feds\n•
\nResearch Initiatives\n•\nacademic\n•\nNGO or watchdog\n•\nCitizen Driven\n•
\ngrassroots efforts\n•\nEnd of Term\n•\nfocused but large-scale
multi-\ninstitutional project\nNovember 14, 2018\nIIPC - WAC 2018\n4\n\nwork
collaboratively to preserve public U.S. Government websites\ndocument federal
agencies' presence on the web during the end of Presidential \nterms\nenhance the
existing research collections of the partner institutions\nraise awareness about
the need for preservation\nengage with researchers and subject experts\ngoals of
the end of term project\nNovember 14, 2018\nIIPC - WAC 2018\n5\n\neot
collaborative distribution of work\n•\nIA: crawling, preservation, access, full-
text \nsearch, metadata, research tool & support\n•\nLC: crawling, preservation,
```

Figure 5: Raw extracted text before cleaning and enhancement

```
---
# **The End of Term Archive: Collaboratively Preserving the United States Government Web**
*Presented by:**
**Mark Phillips**
UNT Libraries
@vphill | mark.phillips@unt.edu
**Abbie Grotke**
Library of Congress
@agrotke | abgr@loc.gov
*Date:** November 14, 2018
*Event:** IIPC - WAC 2018
---
## **1. Introduction and Context**
**Inspiration:** "It all began a long, long, time ago, in a far away place..." (referring to the United States)
[https://flic.kr/p/4N2jHU](https://flic.kr/p/4N2jHU)
[https://flic.kr/p/4JNkLE](https://flic.kr/p/4JNkLE)
**Original End of Term Web Archive Partners:**
Encompassed IIPC & NDIIPP/NDSA partners for the 2008, 2012, and 2016 cycles.
---
## **2. Existing Government Web Archiving Efforts**
*A. Capture, Preservation, & Access Initiatives:**
**Library of Congress (LOC):** Focuses on `.gov` domains, election-related content, and other areas.
**Government Publishing Office (GPO):** Archives agency sites, particularly those with ephemeral content.
**National Archives and Records Administration (NARA):** Conducts biennial web harvests of congressional websites.
**Internet Archive (IA):** Manages global and curated web crawls.
```

Figure 6: Cleaned and AI-enhanced text

14.1.3 Retrieval Method Optimization

The original method selected the top-k chunks directly by similarity score, returning content related to the query but failing to comprehensively cover all relevant aspects. The improved method balances breadth and relevance by considering entire documents rather than isolated chunks.

Previously:

- Computed embedding for the query.
- Used FAISS to retrieve the top-k chunks directly by similarity score.
- Returned those chunks as-is.

Now (after update):

- Still retrieve a larger set of top chunks (e.g., top-30).
- Group chunks by their document ID to see which documents dominate the results.
- Re-rank or filter so that you return chunks from the most relevant documents — rather than just any top individual chunks ,See Figure 9 optmized response.

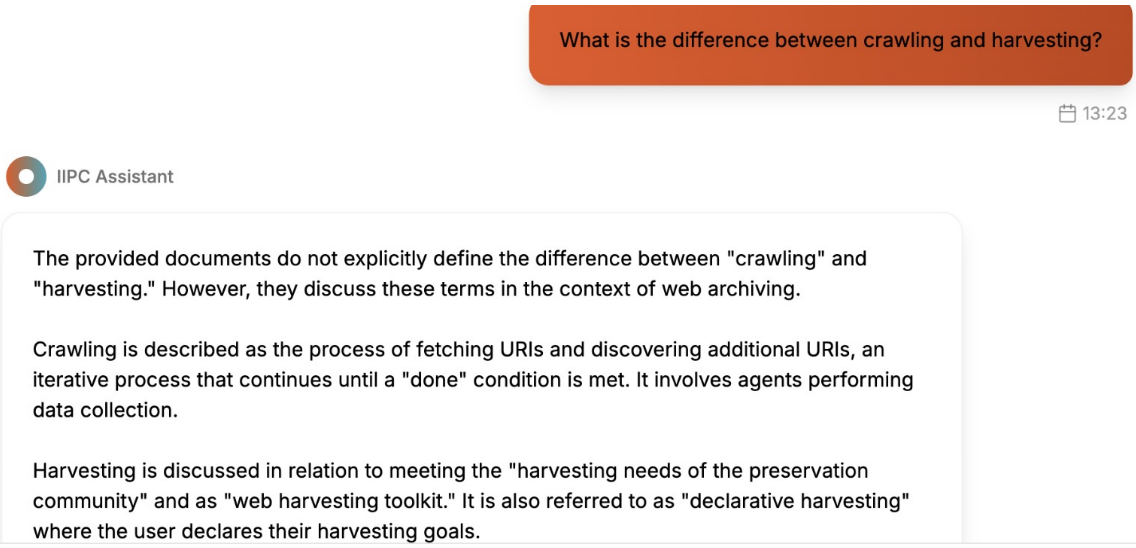

Figure 7: Results using the previous `retrieve_top_k` function and old prompt.

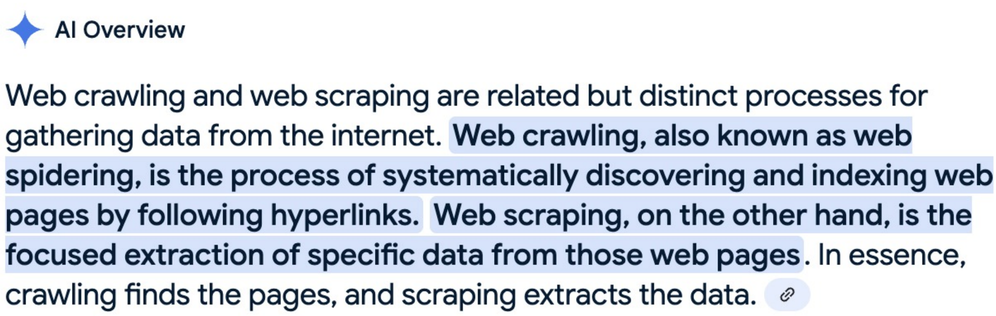

Figure 8: Results from a normal AI tool (baseline).

14.2 System Performance and Improvements

Through iterative improvements in dataset preparation, text enhancement, and retrieval optimization, the system now produces richer, more accurate answers. The prompt was redesigned to align with the domain of web archiving, further improving the quality and relevance of generated responses.

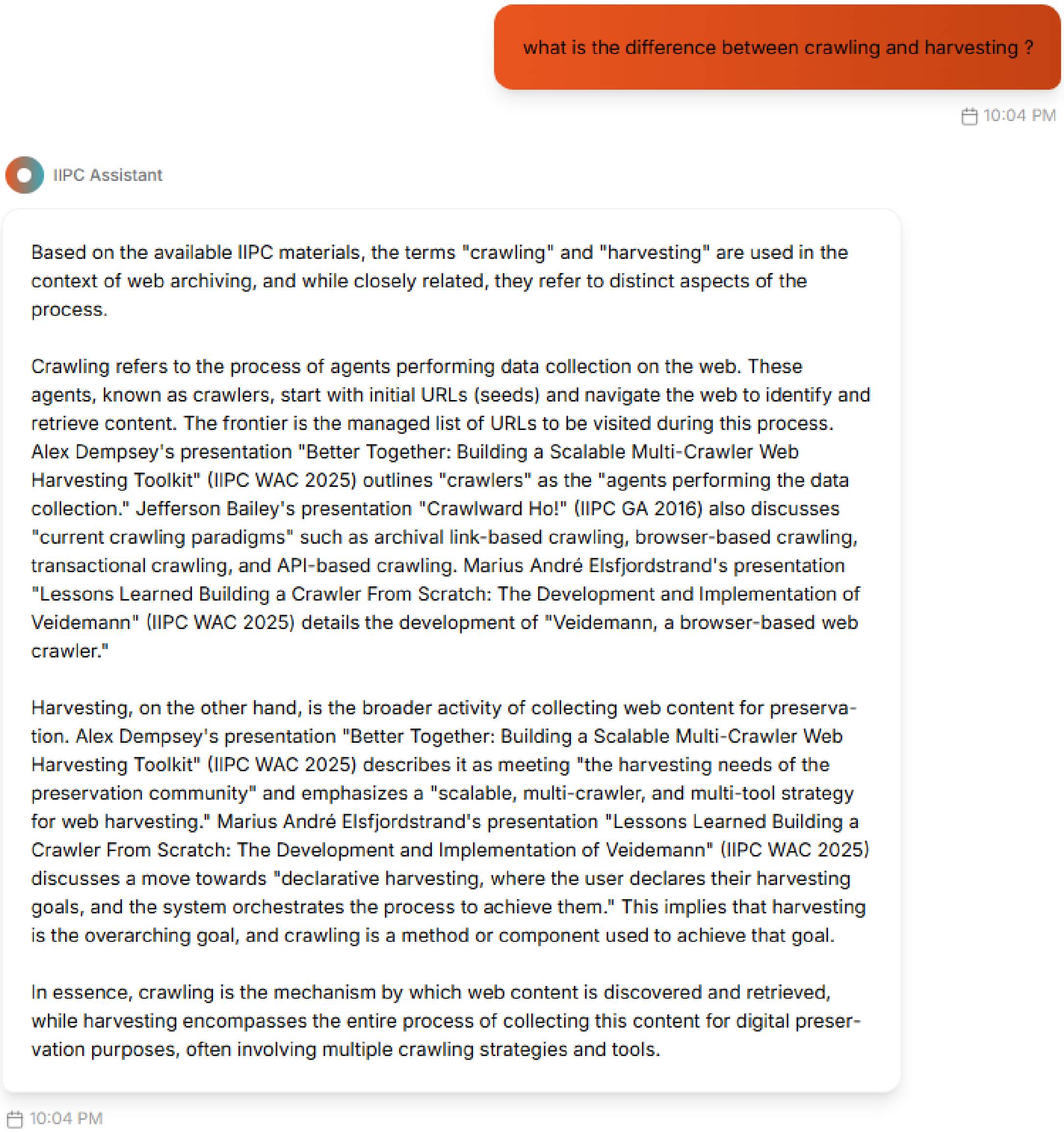

Figure 9: Final enhanced results after applying the updated retrieval function and redesigned prompt.

14.3 Technical Innovation

14.3.1 Data Quality Assurance

- Content Preservation: Maintains original meaning while improving readability
- Metadata Integrity: Complete bibliographic information preservation
- Error Handling: Robust processing of imperfect OCR and extraction results

14.3.2 Scalability Considerations

The system demonstrates efficient handling of large document collections and scalable vector database architecture.

15 Website Screenshots

The website contains four main pages: **Home**, **Browse**, **Chat**, and **About**. The first page below shows the Home and Browse pages together, followed by the Chat and About pages on the next page.

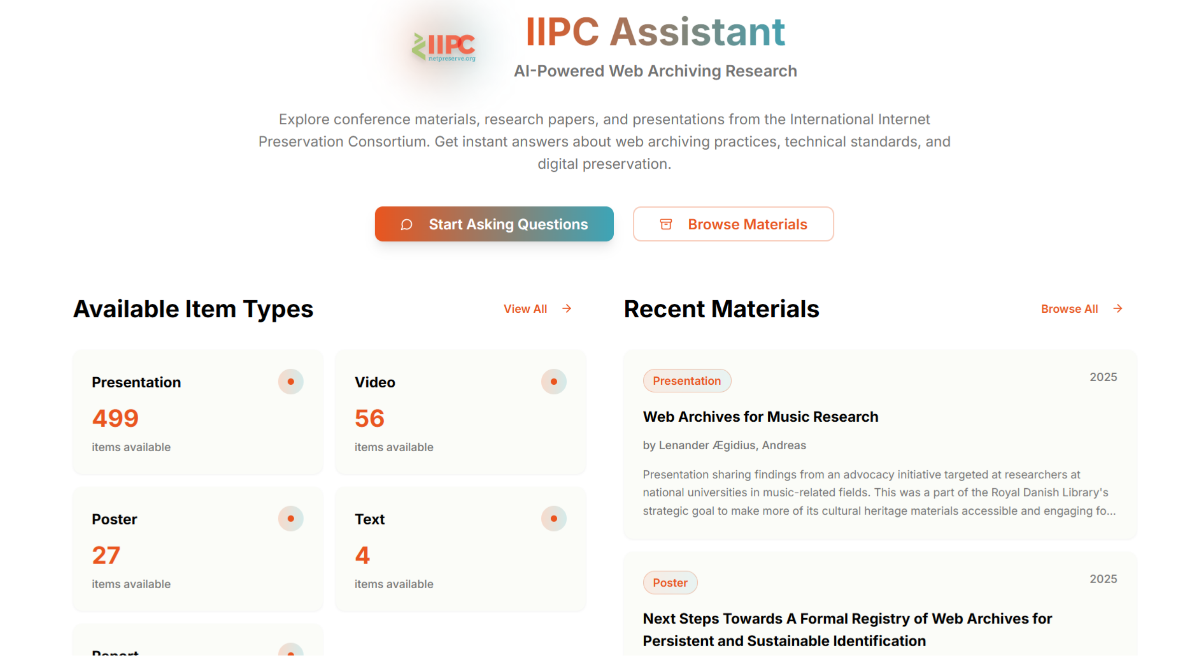

Figure 10: Home page

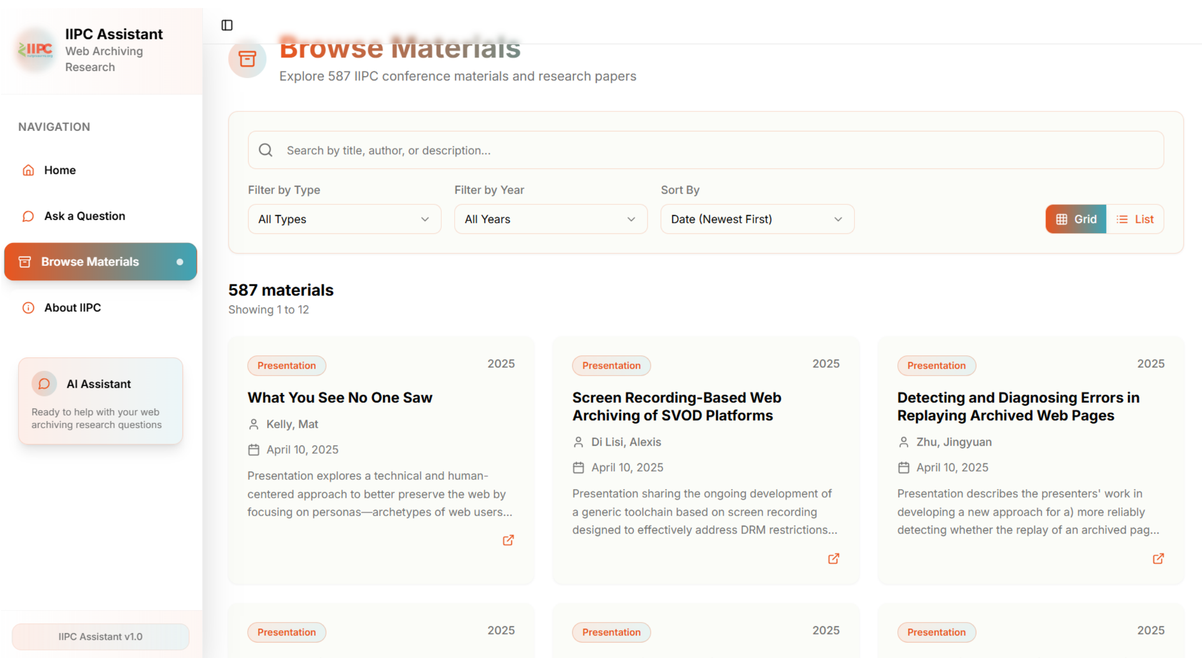

Figure 11: Browse page

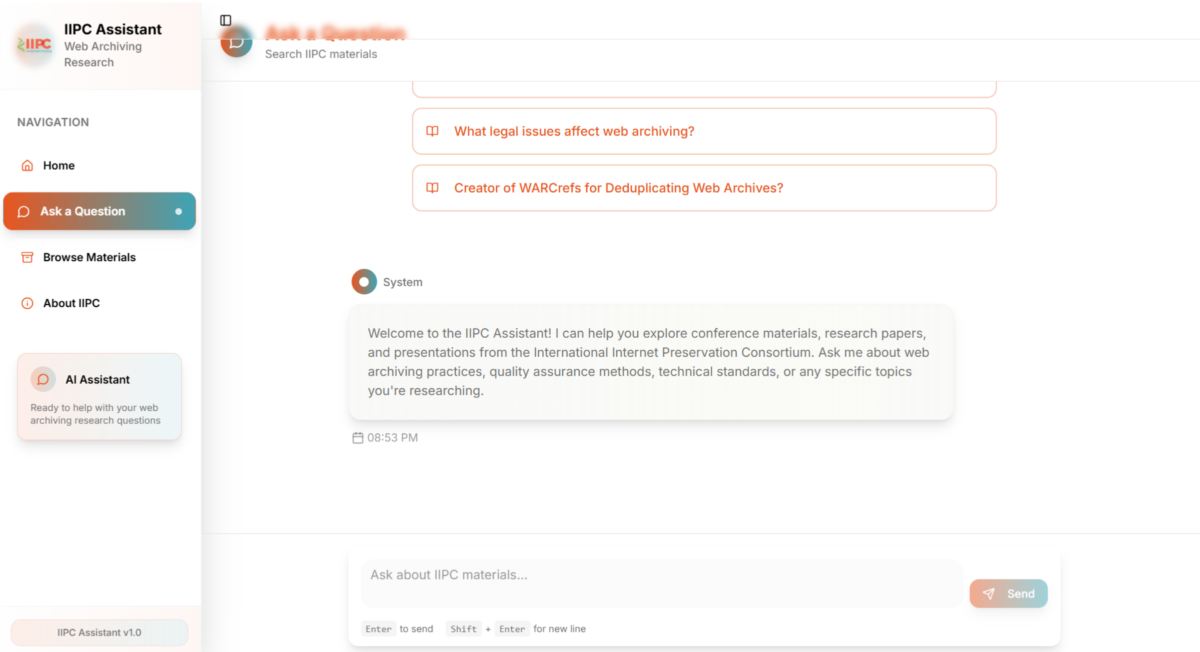

Figure 12: Chat page

Figure 13: About page

16 Conclusion

The IIPC Assistant successfully demonstrates the application of modern AI techniques to enhance accessibility of specialized academic resources. The project achieved its primary objectives of creating an intelligent, searchable interface for IIPC materials through several key innovations:

16.1 Key Achievements

- Successfully harvested and processed 587 IIPC documents with complete metadata preservation
- Implemented advanced text enhancement using AI to transform poorly structured content into coherent, semantically meaningful material
- Developed an optimized retrieval system that balances comprehensiveness with relevance
- Created a responsive web interface supporting real-time interaction across multiple devices

16.2 Technical Contributions

The system represents a practical implementation of Retrieval-Augmented Generation architecture in a specialized domain, demonstrating effective integration of:

- Vector embeddings for semantic search
- Generative AI for contextual response generation
- Modern web technologies for user interaction
- Comprehensive data processing pipelines

References

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Project Resources

- Source Code Repository
- Live Application